

Indian Institute of Technology (ISM), Dhanbad

Department of Computer Science and Engineering
Monsoon End Semester Examination, 2025-26

Examination: BTech Ist Semester
Subject: Computer Programming (NCSV101)

Max. Marks: 50
Time: 2 Hours

Instructions:

- Answer all the questions.
- Use two separate answer booklets, one for each part.
- Write the part number at the top of the front page of each answer booklet.

I have read the instructions mentioned above in this question paper.

Part I

1. Assume that an integer array `arr[]` is already declared and initialized in the program. Write two separate functions, each taking only the array as an argument, to perform the following tasks: (5+5)

A) `int sumEven(int arr[])`

This function must compute and return the sum of all even numbers stored in the array.

B) `float avgOdd(int arr[])`

This function must compute and return the average of all odd numbers stored in the array. If the array contains no odd numbers, the function should return 0.0.

You do not need to write the `main()` function or declare the array. Write only the two function definitions.

2. Find the output of the following code (2.5)

```
#include <stdio.h>
int main() {
    int arr[] = {2, 4, 6, 8, 10};
    int *p = arr;

    printf("%d\n", (*p)++);
    printf("%d\n", *p);
    printf("%d\n", *(++p));
    printf("%d\n", (*p)+2);
    printf("%d\n", *(p+1));

    return 0;
}
```

3. Write a C program to ask the user to enter some integers and store them into an array. Write a recursive function to check whether the elements are in increasing order in the array or not. (1+4)

```
4. struct A {
    int x;
    int y;
};
```

```

struct B {
    double p;
    double q;
};

union U {
    struct A a;      8
    struct B b;      16
    int arr[10];     40
};

union U x

```

Find the size of x.

(2)

5. Write a C program to write the message "Hope you are doing well. Good luck to you." into a file named "example.txt". Then read the file and print the number of words in "example.txt". (2.5+3)

Part II

1. Write your own recursive function to compute a^b . The function should take two integer input parameters for a, b and return the value a^b . You only need to write the recursive function, not the main function. (5)

2. Given two strings txt and pat with their sizes n,m respectively, write a function to find if pat is a substring of txt (a substring is a contiguous sequence of characters within a larger string). If yes, return 1 else return 0. You do not need to write the main function. Use the following function prototype:

```
int checkSubstring(char *txt, int n, char *pat, int m)
```

(8)

3. You are provided with the following code, which takes a positive integer greater than 2 as input and prints the largest prime number smaller than the input:

```

#include <stdio.h>
int before(int n) {
    // Fill your code here
    return -1; // Should not occur for n > 2
}

int main() {
    int n;
    printf("Enter an integer greater than 2: ");
    scanf("%d", &n);
    if (n <= 2) {
        printf("Input must be greater than 2.\n");
        return 1;
    }
    int result = before(n);
    printf("Largest prime smaller than %d is %d\n", n,
result);
    return 0;
}

```

Some part of the code is missing inside the function `before()`. Write the complete `before()` function with the missing code so that the program works as expected. (6)

4. You are provided with the following code, which prints the sequence of some events in the chronological (before to after) order:

```
#include <stdio.h>
#include <string.h>
#define SIZE 4
struct event {
    char title[50];
    int month;
    int year;
};
/* compare(a, b) returns:
   -1 if a is earlier than b
    1 if a is later than b
    0 if equal */
int compare(struct event a, struct event b) {
    // Fill your code here
}
// Function that sorts and prints events
void printOrdered(struct event arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - 1 - i; j++) {
            if (compare(arr[j], arr[j + 1]) > 0) {
                struct event temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    printf("Events from earliest to latest:\n");
    for (int i = 0; i < n; i++) {
        printf("%s (%02d/%d)\n", arr[i].title, arr[i].month, arr[i].year);
    }
}
int main() {
    struct event events[SIZE] = {
        {"Conference", 11, 2025},
        {"Birthday", 4, 2024},
        {"Exam", 6, 2025},
        {"Vacation", 1, 2024}
    };
    printOrdered(events, SIZE);
    return 0;
}
```

Code inside the function `compare()` is missing. Write the complete `compare()` function so that the program works as expected. (6)